

ECA5601 – 5G/ 6G COMMUNICATION

Exp - 9 Massive MIMO Capacity vs SNR and Gain Using MATLAB

Aim:

To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Procedure :

1. Initialize MATLAB and define the Massive MIMO system parameters such as the number of transmit/receive antennas, bandwidth, and SNR range.
2. Generate the MIMO channel and model the received signal considering different SNR values.
3. Calculate the channel capacity for each SNR value using the MIMO capacity equation.
4. Vary the antenna gain and evaluate its effect on the achievable system capacity.
5. Plot capacity versus SNR and gain in MATLAB and analyze how increasing SNR, antenna gain, and the number of antennas improve Massive MIMO capacity.

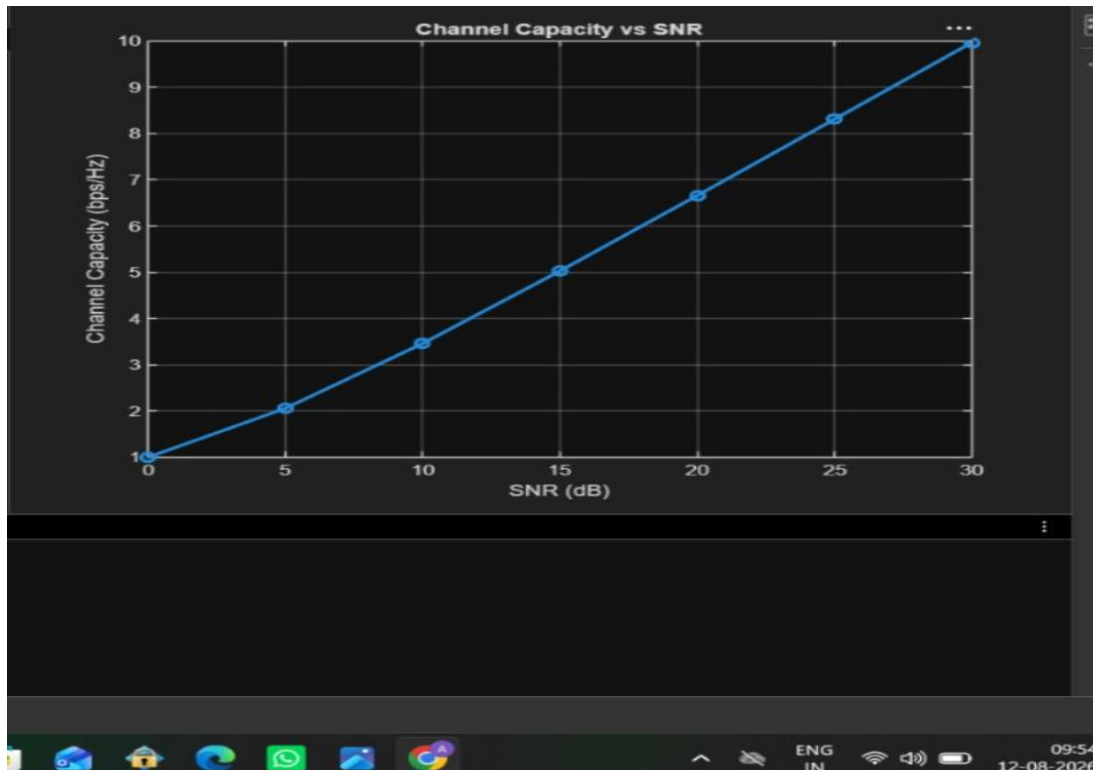
Objective – 1: Analyze Channel Capacity (bps/Hz)

Program :

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
```

```
grid on;  
xlabel('SNR (dB)');  
ylabel('Channel Capacity (bps/Hz)');  
title('Channel Capacity vs SNR');
```

Output :



Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Program :

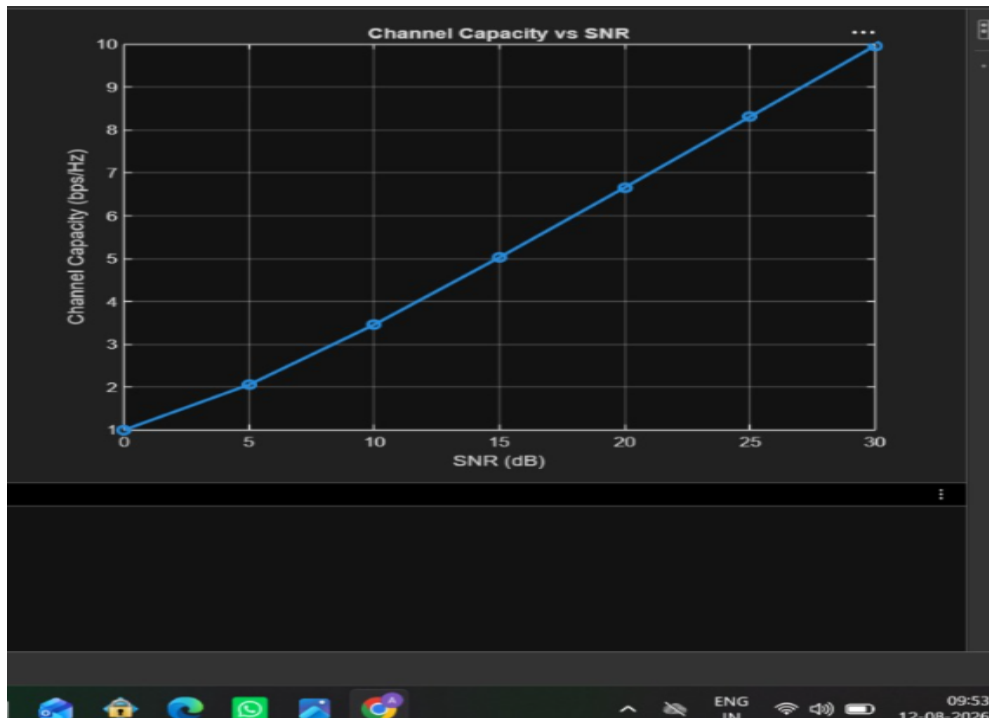
```
clc;  
clear;  
close all;  
snr = 0:5:30;  
snr_linear = 10.^(snr/10);  
capacity = log2(1 + snr_linear);  
figure;  
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
```

```

grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');

```

Output :



Objective – 3: Analyze Antenna Gain (dB)

Program :

```

clc;
clear;
close all;

antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)

figure
bar(antennas, gain)

grid on

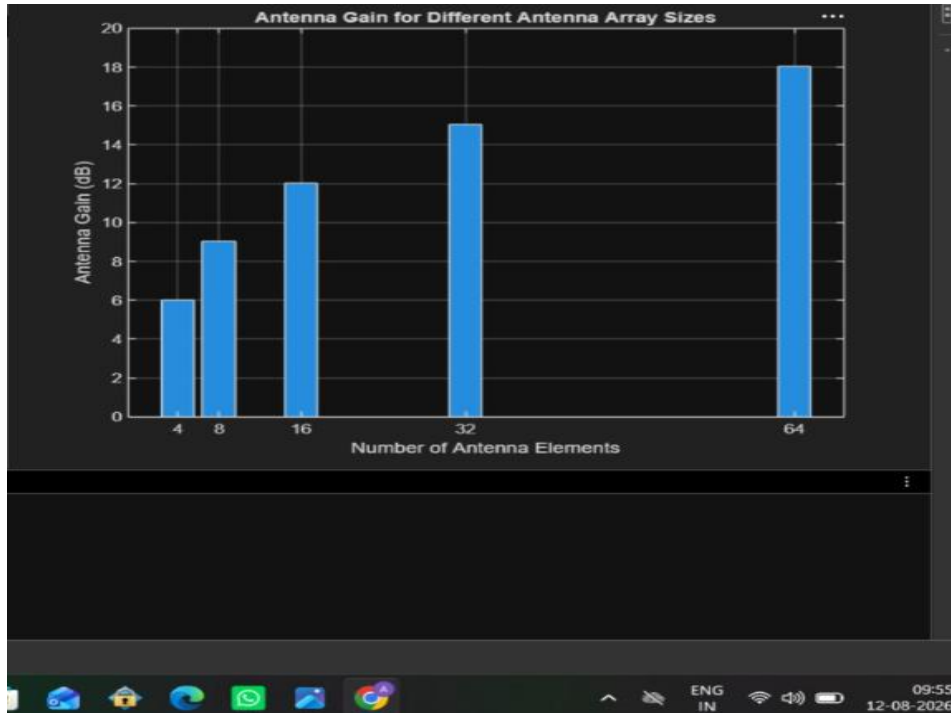
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Antenna Gain (dB)')
```

```
title('Antenna Gain for Different Antenna Array Sizes')
```

Output :



Result :

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.